CA STV MODELING

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1. How it works

Single transferable vote, or STV, is a ranked-choice voting system that can be used to elect one or more winners.[1] In STV, a threshold of first-place votes is required to be elected. If no candidate meets the threshold, the candidate with the fewest first-place votes is eliminated, and their votes are transferred to the next candidate on those ballots. This is repeated until a candidate reaches the threshold; the candidate is elected, and their votes in surplus of the threshold are transferred to the next candidate on those ballots. This is repeated until all seats have been filled.

Here is a simple example of an STV election for two seats.

Example 1. Consider the following 12 cast ballots:

$$\begin{pmatrix} A \\ B \\ C \\ D \\ E \end{pmatrix} \times 5, \begin{pmatrix} B \\ A \\ C \\ D \\ E \end{pmatrix} \times 5, \begin{pmatrix} C \\ B \\ A \\ D \\ E \end{pmatrix} \times 6, \begin{pmatrix} D \\ C \\ B \\ A \\ E \end{pmatrix} \times 3, \begin{pmatrix} E \\ C \\ B \\ A \\ D \end{pmatrix} \times 2.$$

If two seats are up for election, the threshold for election is seven votes (this is the Droop quota and is defined below). The first round of tabulation is as follows:

- A has five votes.
- B has five votes.
- C has six votes.
- D has three votes.
- E has two votes.

Since no one has seven votes, the candidate with the fewest votes is eliminated (E).

Now the ballots are

$$\begin{pmatrix} A \\ B \\ C \\ D \end{pmatrix} \times 5, \begin{pmatrix} B \\ A \\ C \\ D \end{pmatrix} \times 5, \begin{pmatrix} C \\ B \\ A \\ D \end{pmatrix} \times 6, \begin{pmatrix} D \\ C \\ B \\ A \end{pmatrix} \times 3, \begin{pmatrix} C \\ B \\ A \\ D \end{pmatrix} \times 2.$$

The second round of tabulation is as follows:

- A has five votes.
- B has five votes.
- C has eight votes.
- D has three votes.

Since C has eight votes, they are elected and removed from all ballots. C had one surplus vote over threshold, so the ballots with C in first place are discounted by a factor of $\frac{1}{8}$.

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[1] In the case of electing a single seat, STV is often called instant-runoff voting (IRV).

Now the ballots are

$$\begin{pmatrix} A \\ B \\ D \end{pmatrix} \times 5, \begin{pmatrix} B \\ A \\ D \end{pmatrix} \times \frac{6}{8}, \begin{pmatrix} B \\ A \\ D \end{pmatrix} \times 5, \begin{pmatrix} D \\ B \\ A \end{pmatrix} \times 3, \begin{pmatrix} B \\ A \\ D \end{pmatrix} \times \frac{2}{8}.$$

The third round of tabulation is as follows:

- A has five votes.
- B has six votes.
- D has three votes.

Since no one has seven votes, the candidate with the fewest votes is eliminated (D).

Now the ballots are

$$\begin{pmatrix} A \\ B \end{pmatrix} \times 5, \begin{pmatrix} B \\ A \end{pmatrix} \times \frac{6}{8}, \begin{pmatrix} B \\ A \end{pmatrix} \times 5, \begin{pmatrix} B \\ A \end{pmatrix} \times 3, \begin{pmatrix} B \\ A \end{pmatrix} \times \frac{2}{8}.$$

The fourth round of tabulation is as follows:

- A has five votes.
- B has nine votes.

Since B has nine votes, they are elected. Since two candidates have reached threshold, the election is over, no transfer is needed, and the winners are C and B.

In our example, we hid quite a few algorithmic decisions.

Threshold: The first choice to make is the threshold value. Above we used the *Droop quota*, the most common threshold used in STV elections. Let m denote the number of seats to be filled (m stands for *magnitude*). The Droop quota is calculated as

$$\left\lfloor \frac{\#\text{ of votes}}{m+1} + 1 \right\rfloor,$$

where $\lfloor \cdot \rfloor$ denotes rounding down to the nearest integer. One way to interpret the Droop quota is that it is the minimum number of votes such that no more than m candidates can achieve the quota.

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There are other thresholds that can be used, including some which updated dynamically as the tabulation process progresses. For the remainder of this report, we will use the Droop quota.

Transfer method: Another important decision is the method for transferring votes. One option is called *fractional transfer*. In fractional transfer, once a candidate is elected, any ballot with that candidate in first place is discounted by a factor of

$$\frac{\#\text{ of votes} - \text{threshold}}{\#\text{ of votes}},$$

and the vote transfers to the next candidate on the ballot. This is how we transferred votes in our example above.

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In Cambridge, Massachusetts, they use a method called *random transfer*. In random transfer, once a candidate is elected, they randomly select

$$\#\text{ of votes} - \text{threshold}$$

ballots to keep, and the votes are transferred to the next candidate on those ballots. The unselected ballots are no longer used in the tabulation process.

Edge cases: Finally, there are two edge cases that need to be handled. The first is the method for handling multiple candidates crossing the threshold in a single round. Either elect all of them in that round, or elect the one with the most votes, and proceed to another round of tabulation. This decision affects the order in which candidates are elected and can even alter the winner set.

The second consideration is a method for handling ties. Ties can occur whenever two or more candidates have the same number of first-place votes.[2] One method is to just randomly select one of

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the tied candidates; another is to use the first-place vote count from the first round of tabulation.

Taken together, these algorithmic decisions make up an STV election.

2. Background and prior use

2.1. **History in the United States.** Despite sparse implementation today, STV has a long and storied history in the United States. Starting in the early 20th century, STV was adopted for use in local elections across some two-dozen cities, including New York City, Cincinnati, and Cambridge, MA. Almost all of them eventually repealed STV; Cambridge is the notable exception, having used STV continuously since 1941 to elect its nine city council members. See Figure 1 for a timeline of STV's history in the United States. Cite

The story of STV's adoption and repeal in Cinicinatti is emblematic of both the success of this reform strategy, and of the societal friction it created. Like many cities in the United States at the turn of the 20th century, Cinicinatti city hall was a partisan machine; here controlled by the Republicans. The chair of the Hamilton County Republican Committee Bob Cox once said, "The people do the voting. I simply see that the right candidates are elected." In the 1921 election, Republicans won 31 out of 32 seats on the city council, despite Democrats making up about 44% of the electorate.

In response, reformers from the Charter Committee introduced an STV reform measure to the 1924 ballot, which passed with 69% of the vote. By most measures, the reform was a success. The city elected its first Black councillors, with 10 out of 15 councils elected under STV having at least one Black councillor. However, the success of Black candidates caused intense friction with White citizens of Cinicinatti. Starting in the mid-1930s, a wave of repeal measures were intoduced. After the possibility arose that the council would elect a Black mayor, the fifth repeal effort passed in 1957, with White-majority preceincts voting for repeal 2-1, while Black-majority precincts voting 4-1 to keep STV. Cite

The story in New York is largely the same. The implementation of STV in 1937 led to a wave of women, people of color, and minority parties winning seats on the city's board of Aldermen. When two communists were elected in the 1940s, the energy from the Red Scare proved too much and STV was repealed in 1947.

What is illustrated in the history of STV in the United States is that the reform worked as intended. STV, through multi-winner elections and voting thresholds, enabled candidates without majority support to be elected and gave political voice to marginalized groups. Ultimately, this success was its own downfall. Today, we see a renewed interest in STV, with cities like New York City adopting single-seat STV for mayor, Portland, OR adopting STV for city council, and Minneapolis, MN adopting STV for two municipal boards.

2.2. **Use in other countries.**

3. Pros and Cons

3.1. **Pros.**

Flexible: STV can be used with or without districts. It is flexible enough to be used either to elect a single winner per district, to elect multiple winners per district, or to elect all winners at large. This flexibility allows STV to be used in a variety of contexts, from at-large city councils to districted legislative bodies.

Proportional tendencies: STV in the multi-winner setting tends to produce more proportional outcomes than plurality voting. This is in part because STV dampens the power of ballots that have already had a candidate elected, thus giving smaller voting blocs the change to elect a candidate of choice.

[2] This is not an edge case; ties occur in real-world STV elections. See Round 8 in District 1 of the Portland, OR city council election in 2024.

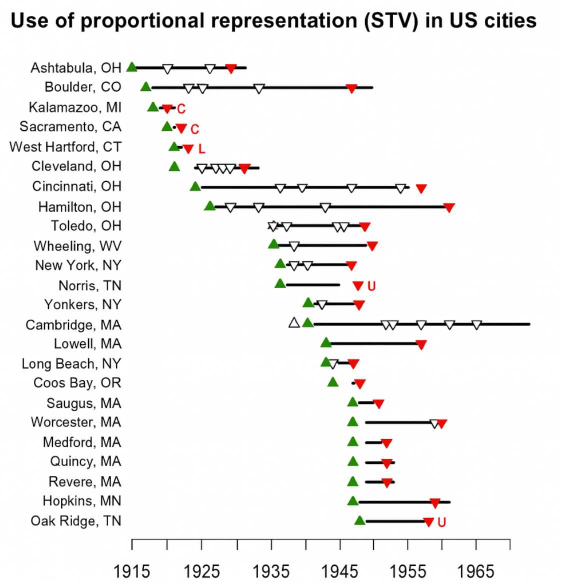

Figure 1. A timeline of STV's history in the United States. Upward green triangles denote adoption of STV, while downward red triangles denote repeal of STV. White triangles denote a failed attempt. Figure source: https://www.voteguy.com/2016/12/30/pr-graphic-for-public-use/

Non-partisan: Unlike some list-based voting systems, STV does not require candidates to run under a party label. This lowers the barrier to entry for candidates who may not enjoy the support of a political party.

3.2. Cons.

Complex mechanism: STV is a relatively complex voting mechanism, and it can be difficult for voters to understand how their vote is converted into an outcome.

Voter burden: As a voting system, STV *technically* requires that each voter provides a complete ballot. That is, each ballot needs to list a candidate in each ranking position. This can be a burden for voters, especially if they are not familiar with the candidates or if they are voting in an election with a long ballot. Of course, many voters do not rank all the candidates and many cities actually cap the ballot length to make it more manageable, and STV still manages to work in the presence of short ballots in practice.

Vote exhaustion: If a voter does not complete their ballot, it is possible that through candidate elimination their vote will no longer be an active part of the tabulation process.

Vote leakage: Votes "for" a party can leak to other parties.

In practice, much of the cons list can be mitigated through voter education and outreach. The mechnism can be explained, and voters can be encouraged to complete their ballots. Cities that have implemented STV have found that mistakes on ballots decrease over time as voters become more familiar with the system.

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CA Single-subject rule concerns

4. Modeling STV in CA

We consider shifting California's lower and upper legislative chambers to STV. There are 80 districts in the lower chamber and 40 districts in the upper chamber. We compare electing 5 members per district with STV to 4 members per district with STV, as well as the current system of single member districts with plurality voting. We identify two different axes to evaluate the impact of the reform along: racial representation and partisan representation. For racial representation we consider splitting the electorate into Hispanic and non-Hispanic voters. For partisan representation we consider splitting the electorate into Democratic and Republican voters.

In order to evaluate the impact of shifting to STV, we need to generate realistic ranked-choice ballots for voters.

4.1. **Generating realistic ranked preferences for voters.** One of the primary challenges with deciding on an electoral reform is that there is not a wealth of historical ranked-choice data available to construct realistic preferences for voters, thus making it difficult to evaluate the impact of the reform. There is a growing body of work in computational social choice that allows us to generate realistic preferences for voters using parameterized mathematical models.

While a full discussion of the mathematical models is beyond the scope of this report, we briefly introduce them here. We use three different models to generate ranked-ballots for voters: the slate-Plackett-Luce sampler, the slate-Bradley-Terry sampler, and the Cambridge sampler. We first assume that voters are divided into two blocs and each bloc has a corresponding slate of candidates. Each bloc of voters has an underlying preference for their slate of candidates over the other slate. Each bloc also has preference for the candidates within their own slate. To create a ballot for a voter, a model first decides the order in which the slates appear on the ballot. Then, for each slate, the model decides the order in which the candidates appear, and fills them in on the ballot in that order.

Consider the following example. Add example

4.2. **Evaluating the impact of the reform.**

Story of the results goes here.

5. Conclusion

Conclusion goes here.